

A lower bound of 3.671741 for the random 3-SAT threshold

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Abstract

Combining the spatial-coupling lower bound on the zero-violation-density threshold with the new linear violation robustness theorem yields

$$\alpha_3 \geq A := \inf_{0 \leq x < 1} \frac{4(1 - x + x \log x)}{(1 - x)^3} > 3.671741.$$

The first input is due to Achlioptas, Hassani, Macris, and Urbanke; the second is Proposition 4.6 of OpenAI's *Computing the Random 3-SAT Threshold*. We give the transfer argument and a finite exact rational certificate for the numerical inequality. The constant A comes from the earlier spatial-coupling analysis; the deduction here transfers that bound to exact satisfiability.

1 Statement and the missing implication

Let $\Phi(n, m)$ have n Boolean variables and m independent clauses, each using three uniformly chosen distinct variables with independent fair signs. Whole clauses may repeat. Let α_3 be its limiting satisfiability threshold, whose existence is established in [4] and also in [3, Theorem 4.4], and let

$$V(F) = \min_{\sigma} \#\{\text{clauses of } F \text{ violated by } \sigma\}.$$

Corollary 1. *Taking the spatial-coupling theorem of [1, 2] and the robustness theorem of [3] as inputs,*

$$\alpha_3 \geq A > \frac{3671741}{10^6}, \quad A = \inf_{0 \leq x < 1} \frac{4(1 - x + x \log x)}{(1 - x)^3},$$

where $0 \log 0 = 0$. In particular, for every fixed $0 \leq a \leq 3671741/10^6$, $\Phi(n, \lfloor an \rfloor)$ is satisfiable with probability tending to one.

Together with the upper-bound theorem of [5], this gives

$$\boxed{3.671741 < \alpha_3 \leq 4.268.}$$

The spatial-coupling result concerns vanishing minimum violation *density*, rather than exact satisfaction. The new robustness proposition supplies the necessary connection. Threshold existence or computability alone is not substituted for that proposition.

2 Two probabilistic inputs

Spatial coupling. Achlioptas, Hassani, Macris, and Urbanke establish a lower bound on the rough threshold: at every fixed density $a < A$, the limiting expected minimum number of violations per variable is zero [1]. Their full manuscript, Theorem 4 and equations (90) and (92), characterizes the coupled unit-clause threshold using

$$\Phi_{\text{ind}}(\ell, a, 3) = 2 - \ell(1 - \log(\ell/2)) - \frac{a}{2}(1 - \ell/2)^3.$$

Setting $\ell = 2x$ gives the expression for A above. The passage to ordinary formulas uses the asymptotic energy interpolation of Hassani, Macris, and Urbanke [2, Theorems 1–2]. One takes the number of variables per position to infinity, then the chain length to infinity at fixed coupling width, and finally the width to infinity. We use this limiting statement, not a finite-chain equality of thresholds.

Linear violation robustness. In the relaxed Poisson model, variable positions inside each clause are independently sampled, and the number of clauses is Poisson with mean an . Proposition 4.6 of the OpenAI preprint [3] states that, for every fixed $a > \alpha_3$, there exists $\varepsilon(a) > 0$ such that

$$\mathbb{P}\{V(F_{n, \text{Poi}(an)}^{\text{rep}}) > \lfloor \varepsilon(a)n \rfloor\} \longrightarrow 1. \quad (1)$$

Theorem 4.4 of that paper identifies the threshold in this model with the proper-clause threshold. The robustness proof uses its free-deletion estimates; it does not assume the spatial-coupling numerical bound or the predicted threshold value.

3 Transferring zero violation density

Adding, deleting, or replacing one clause changes V by at most one. This gives the needed model transfers without an assertion about finite-size satisfiability probabilities.

Write $F_{n,m}^{\text{rep}}$ for fixed-count clauses with independently sampled variable positions, and $F_{n,m}^{\text{prop}}$ for proper clauses. Resample every nonproper clause independently from the proper law. Then, with $m = \lfloor an \rfloor$,

$$|\mathbb{E}V(F_{n,m}^{\text{rep}}) - \mathbb{E}V(F_{n,m}^{\text{prop}})| \leq m[1 - (1 - 1/n)(1 - 2/n)] \leq 3m/n. \quad (2)$$

For an independent $M \sim \text{Poi}(an)$, use one common infinite relaxed clause stream. Cauchy–Schwarz gives

$$|\mathbb{E}V(F_{n,M}^{\text{rep}}) - \mathbb{E}V(F_{n,m}^{\text{rep}})| \leq \mathbb{E}|M - m| \leq \sqrt{an + 1}. \quad (3)$$

After division by n , both errors vanish. Thus the zero-energy conclusion from spatial coupling transfers to the model of (1).

Fix $a < A$. The spatial-coupling conclusion is

$$\frac{1}{n}\mathbb{E}V(F_{n, \text{Poi}(an)}^{\text{rep}}) \longrightarrow 0.$$

If $a > \alpha_3$, equation (1) instead gives

$$\liminf_{n \rightarrow \infty} \frac{1}{n}\mathbb{E}V(F_{n, \text{Poi}(an)}^{\text{rep}}) \geq \varepsilon(a) > 0,$$

a contradiction. Therefore $a \leq \alpha_3$ for every $a < A$, proving $\alpha_3 \geq A$. Only a numerical lower bound for A remains.

4 An exact rational certificate

Set

$$a_0 = \frac{3671741}{10^6}, \quad N = 128, \quad y = \frac{883413964}{10^9}, \quad \eta = \frac{9}{10^8}.$$

For $t = 1 - x \in [0, 1]$, the entropy series is

$$1 - x + x \log x = \sum_{k=2}^{\infty} \frac{t^k}{k(k-1)}.$$

At $t = 1$ the identity follows by continuity or telescoping. Define the rational polynomial

$$P(t) = \sum_{k=2}^{128} \frac{4t^{k-2}}{k(k-1)} - a_0 t.$$

Every coefficient of P'' is nonnegative, so P is convex on $[0, 1]$. Its tangent at y therefore gives, for every $t \in [0, 1]$,

$$P(t) \geq P(y) + P'(y)(t - y) \geq P(y) - |P'(y)|.$$

Exact rational arithmetic proves

$$\boxed{P(y) - |P'(y)| > \frac{9}{10^8}.} \tag{4}$$

The accompanying standard-library Python program `certify_3671741.py` evaluates P and P' both by Horner's rule and by direct summation, and checks (4) using integer comparisons. For orientation only, the rational values are approximately

$$P(y) = 9.3928406844 \cdot 10^{-8}, \quad P'(y) = 3.7588568378 \cdot 10^{-9}.$$

These decimal values are not used by the verifier.

The omitted series terms are nonnegative. Thus for $0 < t \leq 1$,

$$\begin{aligned} \frac{4(1 - x + x \log x)}{(1 - x)^3} - a_0 &= \frac{1}{t} \left(\sum_{k=2}^{\infty} \frac{4t^{k-2}}{k(k-1)} - a_0 t \right) \\ &\geq \frac{P(t)}{t} > \eta. \end{aligned}$$

Taking the infimum proves $A \geq a_0 + \eta > a_0$, completing Corollary 1. In particular the theorem's strict inequality does not depend on interpreting a rounded decimal as an exact bound.

The stationary equation for the interior minimizer is

$$(1 + 2x) \log x + 3(1 - x) = 0.$$

Its numerical solution gives $x \approx 0.11658603275812085$ and $A \approx 3.671741106632344$. These approximations describe the exact variational constant; the rational certificate above is sufficient for the stated lower bound.

5 Attribution and verification

The constant A is inherited from the spatial-coupling work of Achlioptas, Hassani, Macris, and Urbanke. The contribution of this note is the deduction of an exact-satisfiability lower bound using the violation-robustness proposition of [3]. The upper bound 4.268 is a separate result [5] and is not an input to the lower-bound argument.

The probabilistic results are used as cited theorems. The accompanying verifier checks the scalar inequality using exact rational arithmetic; it does not verify the proofs of those theorems. This note has no complete Lean formalization.

Use of language models. OpenAI language models assisted with identifying the combination of results, checking the deduction, developing the arithmetic certificate, and drafting this note. Additional checks were performed in separate model runs; these do not constitute independent human peer review. The author directed the work and is responsible for the final text.

References

- [1] D. Achlioptas, S. H. Hassani, N. Macris, and R. Urbanke. *Bounds for Random Constraint Satisfaction Problems via Spatial Coupling*. SODA 2016, pp. 469–479. doi:10.1137/1.9781611974331.ch35. The authors’ full manuscript is available at <https://di.uoa.gr/~optas/papers/spatial.pdf>; Theorem 4 and equations (90) and (92) are in the appended full version, pp. 29–30.
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